

Fraud Operations: AI Implementation Summary

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Domain: AI & Data Analytics Capstone

1. Model Performance & Selection

Which model performed best and why?

The **Optimized XGBoost Classifier** was the definitive best-performing model. After hyperparameter tuning with Optuna and threshold optimization, it achieved a PR-AUC of **0.6200** and an F1-Score of **0.5952**, vastly outperforming both LightGBM and the unsupervised Isolation Forest. XGBoost excelled because its gradient boosting architecture inherently handles complex, non-linear interactions between hundreds of variables, and it responded exceptionally well to the SMOTE-balanced training data.

2. The Importance of PR-AUC

Why PR-AUC matters more than accuracy in fraud detection:

In our highly imbalanced dataset, the legitimate transactions outnumber fraud by roughly 96.5% to 3.5%. A model that simply guesses "Legitimate" every single time would achieve 96.5% Accuracy, yet be completely useless for stopping fraud.

Precision-Recall AUC (PR-AUC) specifically measures how well the model identifies the *minority* class (fraud) without triggering too many false alarms (Precision) and without missing actual fraud cases (Recall). It is the only metric that reflects true business value in anomaly detection.

3. Explainable AI (SHAP) Insights

Top 3 fraud signals identified by SHAP:

1. **Time of Day (HourOfDay):** The timing of the transaction is the single most critical factor in the model. The wide spread of data points shows that transactions occurring during specific operational windows (represented by the blue dots pushing far to the right) significantly increase the probability of a transaction being flagged as fraud.

2. **Card Type (card6):** The specific type of card used (e.g., credit vs. debit) is the second strongest indicator of risk. The plot shows a severe divide where one specific card type (represented by the blue points on the right) carries a substantially higher fraud risk, while the other type (red points) leans heavily toward legitimate transactions.

3. **Associated Metrics Count (C14):** This anonymized counting feature (which typically represents the number of addresses or specific contact details associated with a card) is a major trigger. The plot reveals a long tail of low values (blue dots) stretching far to

the right, meaning that unusually low counts in this category strongly drive the model to predict fraud.

4. Risk Segmentation Analysis

Common characteristics of Critical Risk transactions:

Transactions flagged in the "Critical Risk" tier (Probability ≥ 0.75) exhibited distinct patterns. The average transaction amount in this tier was approximately **\$107.48**, which is consistently higher or more erratic than the "Clear" tier. Furthermore, these transactions often clustered around specific off-peak hours (e.g., 19:00 / 7:00 PM), suggesting coordinated efforts by bad actors attempting to bypass standard business-hour monitoring.

5. Actionable Business Policies

2 actionable fraud prevention policies:

1. **Dynamic Friction for Critical Tier:** Any transaction scoring above 0.75 probability should trigger immediate Multi-Factor Authentication (MFA) or an SMS verification challenge rather than a hard decline, protecting revenue while stopping fraud.
2. **Velocity Limits on High-Risk Hours:** Implement strict velocity caps (limiting the number of transactions per minute) during the identified high-risk operational windows (e.g., late evenings), specifically for accounts exhibiting high AmtToMeanRatio metrics.

6. Financial Impact & Limitations

Estimated money saved annually:

In our test sample alone, the model successfully isolated hundreds of critical fraud attempts averaging \$107 each. Extrapolating this 3.35% detection rate across an annual volume of millions of transactions, deploying this automated preventative system is estimated to save the business **upwards of \$1.5M to \$2M annually** in chargeback fees and lost inventory.

Model limitations & Future Data:

The primary limitation of this model is its reliance on historical, static features. Fraudsters constantly evolve their tactics (concept drift). To improve future performance, the system should ingest **Additional Data** such as:

- **Behavioral Biometrics:** Tracking mouse movements, typing speed, or screen touches to differentiate bots from humans.

- **Geolocation Data:** IP-to-shipping-address distance metrics to catch identity thieves routing packages to foreign locations.